

Let P be an integral polytope.

$$H(\lambda(P), t) = \frac{h_0^* + h_1^* t + \dots + h_d^* t^d}{(1-t)^{d+1}}$$

Then: $h_i^* \geq 0$

Thm (Hochster): $A(P)$ is Cohen-Macaulay.

Proof sketch: If P is a simplex, every $\beta \in C(P)$ can be written uniquely as

$$\alpha + \sum_{i=1}^{d+1} q_i(v_i, 1) \quad q_i \in \mathbb{Z}_{\geq 0}$$

v_i are the vertices, $\alpha \in S =$ fundamental domain

This implies $\{x^{v_i}y : i=1, \dots, d+1\}$ is a linear system of parameters for $A(P)$. Can check this is a regular sequence. $\Rightarrow$ Cohen-Macaulay.

For general P , take a triangulation of P and use homological algebra.

A **unimodular simplex** in $\mathbb{R}^n$ is a ^{integral} simplex with volume $1/n!$. Alternatively, it is a ^{integral} simplex whose edge directions integrally span $\mathbb{Z}^n$.

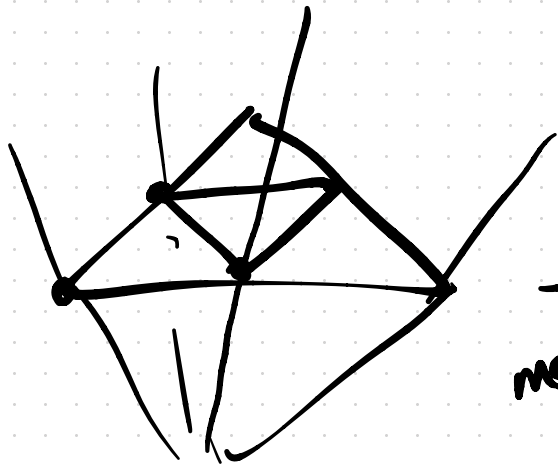
Also defined for not-full-dimensional simplices.

A **unimodular triangulation** is a triangulation with all unimodular simplices.

Prop: If P has a unimodular triangulation Δ , then

$$H(A(P), t) = H(k[\Delta], t)$$

Proof sketch:



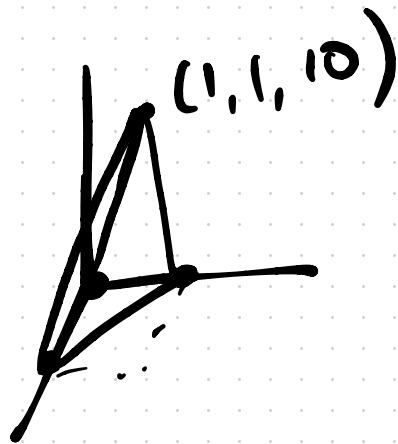
For a unimodular simplex σ

$$A(\sigma) \cong k[x_1, \dots, x_{d+1}]$$
$$\cong k[\sigma]$$

The monomials in $A(P)$ and $k[\Delta]$ are formed by gluing these $A(\sigma)$ together and identifying variables.

Cor: $h_i^*(P) = h_i(\Delta)$ for all i in $\dim \geq 3$

However, not all integral polytopes have unimodular triangulations.



Thm (Hochster): $A(P)$ is Cohen-Macaulay.

Prop: $A(P)$ has a linear system of parameters.

(Not automatic since $A(P) \rightarrow$ not generated in degree one.)

Proof: Every $\beta \in C(P)$ can be written as

$$\alpha + \sum_{i=1}^{d+1} q_i (v_i, 1) \quad \begin{array}{l} q_i \in \mathbb{Z}_{\geq 0} \\ \alpha \in S \end{array}$$

as before. Let $B(P)$ be the subalgebra of $A(P)$ generated by the x^{v_i} 's. $B(P)$ is generated finitely.

in degree one, so by the Noether Normalization Lemma, it has a lsep $\theta_1, \dots, \theta_{d+1}$.

Therefore, there exist $\eta_1, \dots, \eta_r \in B(P)$ such that every element in $B(P)$ is of the form

$$\sum \eta_i p_i(\theta_1, \dots, \theta_{d+1}) \quad p_i \text{ polynomial}$$

Therefore, every element in $A(P)$ is of the form

$$\sum_{\alpha} \sum_i \eta_i x^{\alpha} p_{i,\alpha}(\theta_1, \dots, \theta_{d+1})$$

and there are finitely many $n_i x^{\alpha}$. So

$\theta_1, \dots, \theta_{d+1}$ is a lsp for $A(P)$.

Let P be an integral polytope.

$$H(A(P), t) = \frac{h_0^* + h_1^* t + \dots + h_d^* t^d}{(1-t)^{d+1}}$$

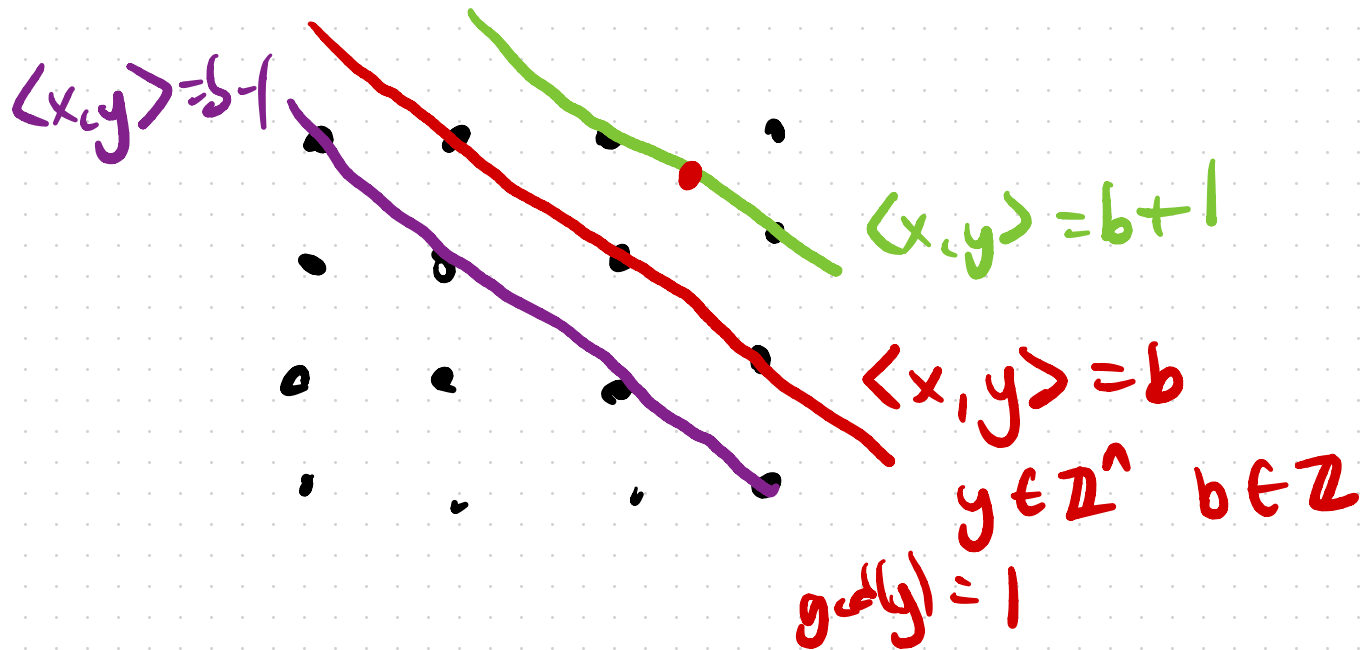
Then: $h_i^* \geq 0$

Proof: Let $\theta_1, \dots, \theta_{d+1}$ be a l.o.p. for $A(P)$.

Because $A(P)$ is Cohen-Macaulay

$$H(A(P)/\langle \theta_1, \dots, \theta_{d+1} \rangle, t) = h_0^* + h_1^* t + \dots + h_d^* t^d.$$

An integral polytope is **reflexive** if it contains a unique lattice point in its relative interior, and this point is **lattice distance** 1 from all its facets.



Alternatively, a reflexive polytope can be translated so it is of the form

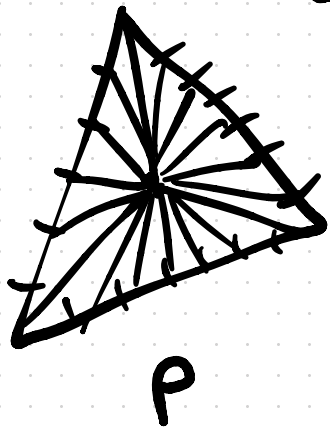
$$P = \{x \in \mathbb{R}^n : \langle x, y_i \rangle \leq 1\} \cap H$$

rational hyperplane
↓
through 0

where $y_i \in \mathbb{Z}^n$ with $\gcd(y_i) = 1$

Thm: $h_i^* = h_{d-i}^*$ if and only if P is reflexive of dim d .

Note: If $2P$ is reflexive and has a unimodular triangulation, then coning over this triangulation with cone point the origin gives



a unimod. triangulation of P whose h -vector is the same as the h -vector of the boundary triangulation, and therefore symmetric.